

High-level statistics Table A3 describes the size and mutation depth of the indel datasets.

Dataset	#Datapoints (Benign/Path.)	Mutation Depth (Min/Mean/Max)	Mutation Source
DMS Assays			
AAV	24,909	1 / 3.57 / 11	randomization
β -Lac	4,751	1 / 1 / 1	library
Kir2.1	10,502	1 / 1.2 / 3	library
MtrA	331	8 / 8 / 8	library
PTEN	314	1 / 1 / 1	library
TP53	341	1 / 1.5 / 2	library
amyloid β	2,354	1 / 14 / 39	library
OCT1	543	1 / 1 / 1	library
Tsuboyama	14,280	1 / 2.7 / 3	library
Assays of Natural and Designed Sequences			
AAV	225,998	3 / 13.9 / 37	model-designed
CM	3,074	1 / 68.9 / 82	model-designed
HIS3	6,102	1 / 8.4 / 29	interpolations between natural sequences
Human Variants			
ClinVar	3k (1,760 / 839)	1 / 1.37 / 3	population variation

Table A3: **Summary of indel datasets.**

A.3.2 Clinical datasets

We collect 65k variants from the ClinVar and gnomAD databases (Table A4).

Dataset	#Proteins	#Variants	#Variants per Protein (Median)
Substitutions	2,525	63k	6
Indels	1,555	3k	1

Table A4: **Summary of ClinVar human variant datasets.**

For our indel benchmark, detailed in Section 4.1, we focus on short indels, less than or equal to three amino acids, which make up over 80% of in-frame indels in our data. There were insufficient benign annotations for indel clinical variants, so gnomAD common variants (allele frequency $> 0.5\%$) were used as a pseudocontrols.

ClinVar processing The clinical substitutions dataset was obtained following the procedure from EVE [Frazer et al., 2021], detailed in Supplementary Methods Section 3 of that paper (which dataset is downloadable from <https://evemodel.org/download/bulk>), but correcting for mapping errors to GRCh38, which yielded 2,525 proteins and 63k variants, with Pathogenic/Likely Pathogenic/Benign/Likely Benign annotations and at least 1 star of clinical evidence - where assertion criteria is provided by a submitter. As a result, our dataset contains significantly more mutants than the dataset from [Frazer et al., 2021] (42k vs. 63k).

The raw set of inframe indels was obtained from ClinVar on February 6th, 2023, by using the following query:

```
("inframe deletion"[Molecular consequence] OR "inframe indel"[Molecular consequence]
OR "inframe insertion"[Molecular consequence])
```

This query yielded 18407 variants. After filtering out invalid/uncertain amino acids, repeats, remaining frameshift variants, and synonymous/stop codons, 17039 (92.5%) remained.

When filtering for Benign/Pathogenic/Likely Benign/Likely Pathogenic annotations (80%+ of annotations are uncertain significance), and selecting variants in genes with at least one P/LP annotation, and filtering indels up to 3 amino acids, $2090 / 18407 = 11.35\%$ of the original variants remained, 330 benign and 1760 pathogenic. When using gnomAD as the benign pseudocontrols, we only keep the 1760 pathogenic variants from ClinVar.

All the preprocessing code from raw ClinVar data is available in the companion codebase.

gnomAD processing The Genome Aggregation Database (gnomAD) [Karczewski et al., 2020] seeks to aggregate genome and exome sequencing data from multiple large-scale sequencing projects, and publishes summary data such as variant allele frequencies in a consistent format. The gnomAD v2.1.1 GRCh38 liftover was downloaded on February 8th 2023 and contains 125,748 exomes and 15,708 genomes. v2 was originally based on the GRCh37 reference sequence and v2.1.1 was lifted over to the GRCh38 reference sequence.

The inframe indels were similarly preprocessed to the ClinVar indels (the preprocessing code from raw data is also available in the repository), yielding 839 common indels up to 3 amino acids in length.

A.3.3 Access

The following provides more details on the code and data Resources (§ 6) accompanying this paper.

The open-source codebase containing a framework for scoring all the benchmarks is available via our GitHub repository at: <https://github.com/OATML-Markslab/ProteinGym>. Modifications of certain baselines (e.g. scoring of long sequences beyond the context size in the ESM suite, or pre-processing of MSAs in MSA Transformer) are also released, and all of the model predictions can be reproduced using this repository. We also include preprocessing code for the clinical data (ClinVar/gnomAD) and DMS assays for reproducibility.

We developed a user-friendly website, <https://www.proteingym.org> containing a leaderboard, detailed results per assay, as well as drill analyses across various dimensions (e.g. mutational depth, taxa).

The DMS assays, model scores, Multiple Sequence Alignments, predicted 3D structures, processed ClinVar/gnomAD datasets, and raw files before preprocessing can all be downloaded from our servers (see download instructions on our GitHub repository). Some model checkpoints and other files necessary for scoring (for baselines such as profileHMM, PROVEAN) are also available via our servers, although most model checkpoints such as ESM-1v are available from their respective repositories.

A.3.4 License

The codebase is open source under the MIT license.

A.4 Baselines

Unless otherwise specified, model scores are calculated by taking the log-ratio of the sequence probabilities between the mutant and wild-type sequences $\log \frac{p(\mathbf{x}_{\text{mut}})}{p(\mathbf{x}_{\text{wt}})}$, following the convention in Hopf et al. [2017].

A.4.1 Zero-shot baselines

Alignment-based models

- **Site-independent Model** We use a site-wise maximum entropy model to infer the contribution of site-specific amino acid constraints without considering explicit epistatic constraints. This model is implemented as referred to in Hopf et al. [2017].
- **HMM** We use the profile hidden Markov model (HMM) implementation in HMMER [Eddy, 2011]. Profile HMMs are frequently used to generate multiple sequence alignments, but also produce log probabilities of sequences that can be used as estimates of fitness for both substitutions and indels [Durbin et al., 1998].
- **EVMutation** EVMutation [Hopf et al., 2017] models pairwise evolutionary couplings between protein sequences using a Potts model (otherwise known as a Markov Random Field).
- **DeepSequence** DeepSequence [Riesselman et al., 2018] uses a VAE architecture to learn higher-order non-linear evolutionary constraints within each protein family. Mutation effect scores are calculated similarly as EVMutation, as the log-ratio between the mutant and wild-type sequence probabilities $\log \frac{p(\mathbf{x}_{\text{mut}}|\theta)}{p(\mathbf{x}_{\text{wt}}|\theta)}$, but using the VAE evidence lower bound (ELBO) as a proxy for $p(\mathbf{x}|\theta)$.
- **WaveNet** We use a previously published dilated convolutional neural network (diCNN) based on the WaveNet architecture [Shin et al., 2021] as an example of a family-specific sequence decoder capable of handling indels. Due to the expense of training a separate model for each protein, we only evaluate this model against the DMS datasets. Sequence scores are calculated as the difference in (length-normalized) log-likelihoods between the mutant and wild-type sequences.
- **EVE** EVE [Frazer et al., 2021] is a Bayesian VAE model architecture for predicting clinical variant effects. The model includes a Gaussian Mixture Model fitted to the background distribution of mutations, in order to provide interpretable protein-specific pathogenicity scores. We use the ClinVar preprocessing pipeline from EVE, and EVE is also used in TranceptEVE [Notin et al., 2022b].

- **GEMME** GEMME is the Global Epistatic Model for predicting Mutational Effects. It infers the conservation of combinations of mutations across the entire sequence according to an evolutionary tree [Engelen et al., 2009] and combines it with site-wise frequencies to calculate a combined epistatic sequence score for mutations [Laine et al., 2019]. GEMME intakes multiple sequence alignments of protein families as well as specific mutations to generate scores. To obtain scores, we used the GEMME web-tool hosted at <http://www.lcqb.upmc.fr/GEMME/submit.html> with default parameters.

Protein language models Protein language models are so called because they all use variants of the Transformer [Vaswani et al., 2017] architecture popularised in natural language processing.

- **UniRep** UniRep [Alley et al., 2019b] trains a Long Short-Term Memory (LSTM) model on UniRef50 [Suzek et al., 2015] sequences. It learns how to internally represent proteins by being trained on next amino acid prediction through minimizing cross-entropy loss. While the core model is trained on unaligned sequences, UniRep can also be fine-tuned on sets of homologous sequences from a given family, retrieved with a MSA. This process is called ‘evotuning’ and typically leads to stronger fitness prediction performance.
- **ESM** ESM-1b [Rives et al., 2021] and ESM-1v [Meier et al., 2021] are protein language models with a Transformer encoder architecture similar to BERT [Devlin et al., 2019] and trained with a Masked-Language Modeling (MLM) objective on UniRef50 and UniRef90 respectively. We extend the original ESM codebase for these two models to handle sequences that are longer than the model context window (ie., 1023 amino acids), with the approach described in Brandes et al. [2023] for ESM-1b and in Notin et al. [2022a] for ESM-1v. We predict fitness for ESM models with the masked-marginal approach introduced in Meier et al. [2021], which provides optimal performance on substitutions, but does not support indels.
- **CARP** CARP [Yang et al., 2023a] is a protein language model trained with a MLM objective on UniRef50. The architecture leverages convolutions instead of self-attention, leading to computational speedups while maintaining high downstream task performance.
- **RITA** RITA [Hesslow et al., 2022] is an autoregressive language model akin to GPT2 [Radford et al., 2019], trained on UniRef100 [Suzek et al., 2015]. Four model sizes are available, ranging from 85 million to 1.2 billion parameters. RITA takes unaligned sequences as input, and can score both substitution and indel mutations.
- **ProGen2** ProGen2 [Nijkamp et al., 2022] is an autoregressive protein language model trained on a mixture of UniRef90 [Suzek et al., 2014] and BFD30 [Steinegger and Söding, 2018]. It follows the standard transformer decoder architecture, and five models of different sizes are available, ranging from 151 million to 6.4 billion parameters. ProGen2 takes unaligned sequences as input, and can score both indel and substitution mutations.
- **VESPA** VESPA [Marquet et al., 2022] is as Single Amino acid Variant (SAV) effect predictor based on a combination the embeddings from the protein language model ProtT5 [Elnaggar et al., 2021], as well as per-residue conservation predictions.

Inverse Folding models Inverse folding models learn the conditional distribution of sequences that are likely to fold to an input protein structure [Ingraham et al., 2019]. Given that there may not be experimentally solved structures for the target sequence of all DMS assays in ProteinGym, we generate input structures using AlphaFold2 (AF2) [Jumper et al., 2021]. The inverse folding model in combination with AF2 encompasses an end-to-end scoring pipeline that only requires a protein sequence to score variants. As the sequence representation size is defined by the size of the input structure, the models we benchmark here can only score substitutions.

- **ProteinMPNN** ProteinMPNN [Dauparas et al., 2022] takes in a protein backbone structure and featurizes it as a graph where backbone (N,C,C α) atoms are nodes and edges are determined via euclidian distance cut-offs. The model uses a message passing neural network (MPNN) [Ingraham et al., 2019] to encode the structure into a latent graph representation. The model then decodes the representation and samples sequences autoregressively.
- **MIF** The Masked Inverse Folding (MIF) and Masked Inverse Folding with Sequence Transfer (MIF-ST) models [Yang et al., 2023b] are structured-conditioned protein language models trained with a MLM objective. MIF is trained on CATH4.2 [Dawson et al., 2016], and MIF-ST further augments the MIF model with embeddings from CARP (640M).
- **ESM-IF1** ESM-IF1 [Hsu et al., 2022b] functions similarly to ProteinMPNN but leverages a Geometric Vector Perceptron [Jing et al., 2020] (an equivariant message passing module ideal for coordinate data) as the architecture for the structure encoder and sequence decoder.

Hybrid models

- **MSA Transformer** The MSA Transformer [Rao et al., 2021] learns a representation of Multiple Sequence Alignments (MSAs) by training an Axial transformer-based transformer [Ho et al., 2019a] with a MLM objective across a diverse set of 26 million MSAs.
- **Tranception** Tranception [Notin et al., 2022a] combines an autoregressive protein language model with inference-time retrieval from a MSA. We evaluate Tranception Small (85M), Tranception Medium (300M parameters) and Tranception Large (700M parameters) both with and without MSA retrieval. Tranception can score both indel and substitution mutations.
- **TranceptEVE** TranceptEVE augments Tranception with priors for the amino acid distribution at each position based on an ensemble of EVE models for the protein family of interest. The final output log probability is thus a weighted sum of that EVE log prior, the log probability from the autoregressive transformer model in Tranception, as well as site-specific log probabilities obtained from a retrieved MSA (as in the inference-time retrieval procedure described in Tranception). TranceptEVE can score both indels and substitutions.

A.4.2 Supervised baselines

We leverage the various supervised baselines defined in Notin et al. [2023]:

- **One-hot encoding (OHE) models** OHE baselines take as input a one-hot encoding representation of the amino acid sequence, together with zero-shot fitness predictions obtained with several of the baselines discussed above in Appendix A.4.1. Both are input into a L2-penalized regression, following the approach discussed in [Hsu et al., 2022a];
- **Embeddings models** Embeddings models are based on mean-pooled embeddings from various protein language models introduced above (e.g., Tranception, ESM-1v, MSA Transformer), augmented with zero-shot fitness predictions from the same model. We refer to Notin et al. [2023] for all implementation details;
- **ProteinNPT** ProteinNPT [Notin et al., 2023] is a semi-supervised non-parametric transformer [Kossen et al., 2022] which learns a joint representation of full batches of labeled sequences. It is trained with a hybrid objective consisting of fitness prediction and masked amino acids reconstruction. The model can be used to predict mutation effects for single or multiple properties simultaneously, and sample novel sequences conditioned on label values of interest.

A.4.3 Clinical baselines

We leverage a set of clinical variant effect predictors from dbNSFP v4.4a [Liu et al., 2011, 2020], which is a database of functional predictions for all possible non-synonymous single-nucleotide variants (nsSNVs) in the human genome.

These models were developed primarily to assess the effects of mutations in humans and are included in clinical benchmarks only:

- **Supervised** The following assays used ClinVar label annotations in their training (or are meta-predictors that contain one or more supervised models): ClinPred [Alirezaie et al., 2018], MetaRNN [Li et al., 2022], BayesDel [Feng, 2017], VEST4 (variant effect scoring tool 4.0) [Carter et al., 2013], REVEL [Ioannidis et al., 2016], VARITY [Wu et al., 2021], gMVP [Zhang et al., 2022], CADD [Rentzsch et al., 2019], PolyPhen2 [Adzhubei et al., 2010], DEOGEN2 [Raimondi et al., 2017], MPC [Samocha et al., 2017], MutationTaster [Schwarz et al., 2010], DANN [Quang et al., 2015], FATHMM[Shihab et al., 2013]
- **Unsupervised** In addition to TranceptEVE, GEMME, EVE and ESM-1b (zero-shot baselines mentioned above), the following unsupervised clinical variant effect predictors were used as baselines: PROVEAN [Choi et al., 2012], SIFT [Ng and Henikoff, 2002], MutationAssessor [Reva et al., 2011], MutPred [Li et al., 2009], PrimateAI [Sundaram et al., 2018], LIST-S2 [Malhis et al., 2020], and LRT [Chun and Fay, 2009].

For ESM-1b, we downloaded precomputed scores from [Brandes and Ntranos, 2023] from a recent study that extended ESM-1b to predict all possible missense variant effects in the human genome [Brandes et al., 2023]. TranceptEVE and EVE models were trained for the subset of 2,525 proteins in the clinical benchmark, and the model weights/scores are provided online for further analysis (See Section A.3.3). GEMME scores were obtained as detailed above. We provide an analysis of performance on clinical datasets vs the subset of assays on human proteins in Fig. 2.

A.5 Detailed performance results

A.5.1 DMS substitution benchmarks

Zero-shot Table A5 shows the results for our zero-shot DMS substitutions benchmark. We report Spearman’s rank correlations and bootstrapped standard error estimates for forty baseline models. Table A6 breaks down our substitution DMS by MSA depth, , Table A7 by function type, Table A8 by taxa, and Table A9 by mutational depth. To compute the final Spearman’s rank correlation reported in Table A5, we first average all the assays for a particular function type together, resulting in five average values (one each for Activity, Binding, Expression, Organismal Fitness, and Stability). The average of these five numbers is the final reported value.

Clustering zero-shot substitution models We clustered the zero-shot models using hierarchical clustering on the vector of NDCG metrics for each dataset in the DMS substitutions (Fig. A1). We find that models with the same architecture tend to cluster together (e.g., RITA models), however, there are exceptions (e.g., ESM-2 models). We also observe that the alignment-based models tend to cluster together, suggesting that training on the same MSA may promote similar scoring behavior.

Supervised Table A10 shows the results for our supervised DMS substitutions benchmark. We report Spearman’s rank correlations for 10 baseline models.

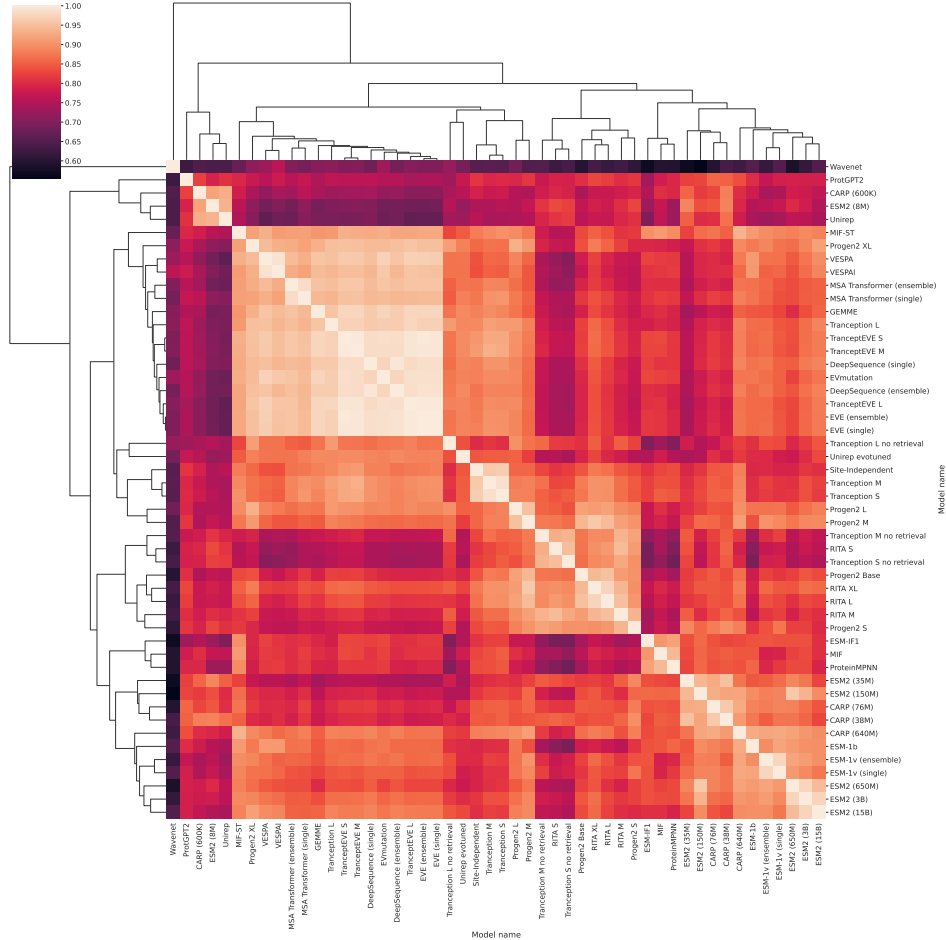


Figure A1: **Hierarchical clustering of zero-shot models by NDCG performance** Heatmap colored by the Pearson correlation of the NDCG@10% values for each DMS assay for each pair of zero-shot models. Lighter color corresponds to higher correlation. The ordering and dendrogram were produced by hierarchical clustering of the correlation values.